

**MATHEMATICAL METHODS IN BIOENGINEERING**  
**TEST # 7**

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Random samples of 50 college students, 40 college faculty, and 60 bankers yielded the following data relating to the numbers of smokers in these samples:

| Group            | Number who smoke |
|------------------|------------------|
| College students | 22               |
| College faculty  | 9                |
| Bankers          | 25               |

- (a) Test the hypothesis, at the 10 percent level of significance, that the same percentages of college students, college faculty, and bankers are smokers.
- (b) Repeat part (a) at the 5 percent level of significance.

**Sol: Chi-Square Test for Homogeneity:** Let  $H_0$  be the null hypothesis that the proportion of smokers is the same across all three populations, versus  $H_a$  that the proportions are not all equal.

First, we construct the full contingency table to find marginal totals ( $n = 150$ ).

| Group            | Smokers      | Non-Smokers  | Total |
|------------------|--------------|--------------|-------|
| College students | 22 (18.6667) | 28 (31.3333) | 50    |
| College faculty  | 9 (14.9333)  | 31 (25.0667) | 40    |
| Bankers          | 25 (22.4000) | 35 (37.6000) | 60    |
| Total            | 56           | 94           | 150   |

*Note: Expected counts  $E_{ij} = \frac{\text{Row Total} \times \text{Column Total}}{n}$  are written in parentheses.*

Using the test statistic formula  $\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$ :

$$\begin{aligned}
 \chi^2 &= \frac{(22 - 18.6667)^2}{18.6667} + \frac{(28 - 31.3333)^2}{31.3333} + \frac{(9 - 14.9333)^2}{14.9333} \\
 &\quad + \frac{(31 - 25.0667)^2}{25.0667} + \frac{(25 - 22.4)^2}{22.4} + \frac{(35 - 37.6)^2}{37.6} \\
 \chi^2 &= \frac{(3.3333)^2}{18.6667} + \frac{(-3.3333)^2}{31.3333} + \frac{(-5.9333)^2}{14.9333} + \frac{(5.9333)^2}{25.0667} + \frac{(2.6)^2}{22.4} + \frac{(-2.6)^2}{37.6} \\
 &= 0.5952 + 0.3546 + 2.3574 + 1.4045 + 0.3018 + 0.1798 \\
 &= 5.1933
 \end{aligned}$$

The degrees of freedom are:

$$df = (3 - 1) \times (2 - 1) = 2$$

**(a) At the 10% significance level ( $\alpha = 0.10$ ):** The critical value is  $\chi_{0.10,2}^2 = 4.605$ . Since our test statistic  $\chi^2 = 5.1933 > 4.605$ , we **reject** the null hypothesis. There is sufficient evidence at this level to conclude that the percentages of smokers are different.

(b) **At the 5% significance level ( $\alpha = 0.05$ ):** The critical value is  $\chi^2_{0.05,2} = 5.991$ . Since  $\chi^2 = 5.1933 \leq 5.991$ , we **fail to reject** the null hypothesis.